

| Question |     |       |  | Marking details                                                                                                                                                                                                             | Marks available |          |          |          |          |          |
|----------|-----|-------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |     |       |  |                                                                                                                                                                                                                             | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
| 4        | (a) | (i)   |  | {Its/gas} {cost/bill} [accept: its cost] decreased [after solar panels installed] [accept: less gas used].<br>Not: electricity didn't go down / decreased [neutral]                                                         |                 |          | 1        | 1        |          |          |
|          |     | (ii)  |  | Gas cost dropped by £950 - £840 [= £110] (1)<br>payback period = $\frac{4\,680}{110} = 42.5$ [y] (1) [Accept 43; not 42]<br><b>Accept</b><br>Total cost dropped by £60 (1)<br>payback period = $\frac{4\,680}{60} = 78$ (1) |                 | 1<br>1   |          | 2        | 2        |          |
|          |     | (iii) |  | Payback time reduced (1)....<br>...[because ] money saved would increase (1)<br>The unit cost of electricity would make no difference to the payback time (1)                                                               |                 | 3        |          | 3        |          |          |
|          | (b) |       |  | Saving = £50 (1)<br>Division of a sum of money by 300 shown (1)<br>Answer 16.7 [p] (1) [accept 17 p, not 16.6 p]<br><br>NB. Answer = 0.16[7] [p] → 2 marks                                                                  | 1               | 2        |          | 3        | 3        |          |
|          |     |       |  | <b>Question 4 total</b>                                                                                                                                                                                                     | <b>1</b>        | <b>7</b> | <b>1</b> | <b>9</b> | <b>5</b> | <b>0</b> |